

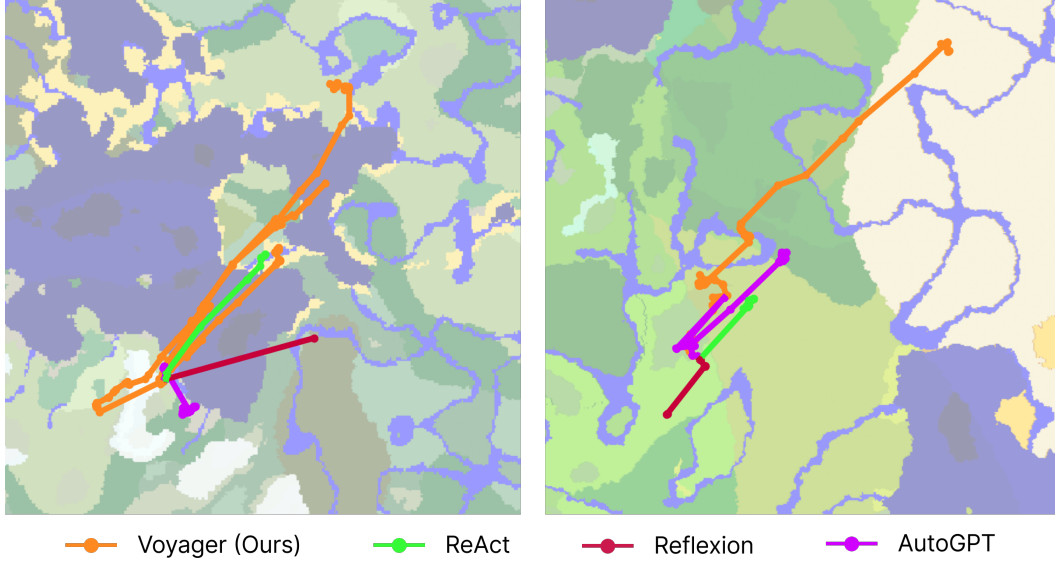


Figure A.2: Map coverage: Two bird’s eye views of Minecraft maps. VOYAGER is able to traverse $2.3\times$ longer distances compared to baselines while crossing diverse terrains. Trajectories are plotted based on the positions where each agent interacts with GPT-4.

B.4.2 Extensive Map Traversal

Agent trajectories for map coverage are displayed in Fig. A.2. Fig. 7 is plotted based on Fig. A.2 by drawing the smallest circle enclosing each trajectory. The terrains traversed by VOYAGER in each trial is

- **Trial 1:** ‘meadow’, ‘desert’, ‘river’, ‘savanna’, ‘forest’, ‘plains’, ‘bamboo_jungle’, ‘dripstone_caves’;
- **Trial 2:** ‘snowy_plains’, ‘frozen_river’, ‘dripstone_caves’, ‘snowy_taiga’, ‘beach’;
- **Trial 3:** ‘flower_forest’, ‘meadow’, ‘old_growth_birch_forest’, ‘snowy_slopes’, ‘frozen_peaks’, ‘forest’, ‘river’, ‘beach’, ‘ocean’, ‘sunflower_plains’, ‘plains’, ‘stony_shore’.

The terrains traversed by ReAct [29] in each trial is

- **Trial 1:** ‘plains’, ‘desert’, ‘jungle’;
- **Trial 2:** ‘snowy_plains’, ‘snowy_taiga’, ‘snowy_slopes’;
- **Trial 3:** ‘dark_forest’, ‘dripstone_caves’, ‘grove’, ‘jagged_peaks’.

The terrains traversed by Reflexion [30] in each trial is

- **Trial 1:** ‘plains’, ‘flower_forest’;
- **Trial 2:** ‘snowy_taiga’;
- **Trial 3:** ‘old_growth_birch_forest’, ‘river’, ‘ocean’, ‘beach’, ‘plains’.

The terrains traversed by AutoGPT [28] in each trial is

- **Trial 1:** ‘plains’, ‘dripstone_caves’, ‘savanna’, ‘meadow’;
- **Trial 2:** ‘snowy_taiga’;
- **Trial 3:** ‘plains’, ‘stony_shore’, ‘forest’, ‘ocean’.

B.4.3 Efficient Zero-Shot Generalization to Unseen Tasks

The results of zero-shot generalization to unseen tasks for the other two tasks are presented in Fig. A.3. Similar to Fig. 8, VOYAGER consistently solves all tasks, while the baselines are unable to solve any

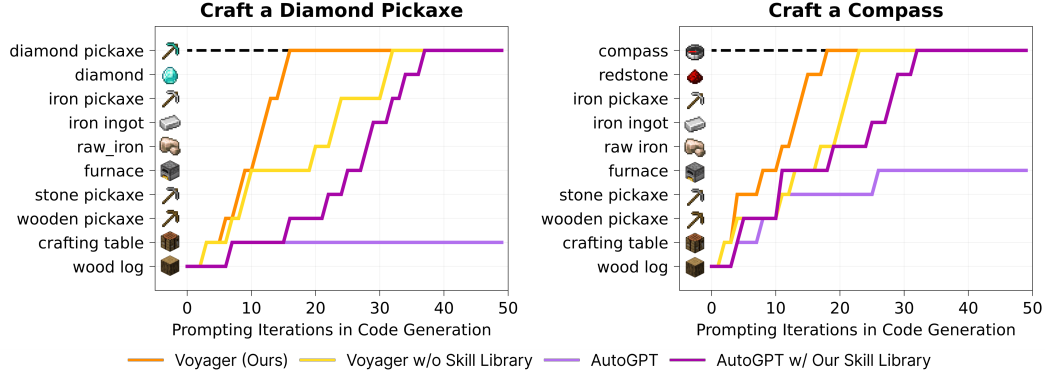


Figure A.3: Zero-shot generalization to unseen tasks. We visualize the intermediate progress of each method on the other two tasks. We do not plot ReAct and Reflexion since they do not make any meaningful progress.

task within 50 prompting iterations. Our skill library, constructed from lifelong learning, not only enhances VOYAGER’s performance but also provides a boost to AutoGPT [28].

B.4.4 Accurate Skill Retrieval

We conduct an evaluation of our skill retrieval (309 samples in total) and the results are in Table. A.4. The top-5 accuracy standing at 96.5% suggests our retrieval process is reliable (note that we include the top-5 relevant skills in the prompt for synthesizing a new skill).

Table A.4: Skill retrieval accuracy.

Top-1 Acc	Top-2 Acc	Top-3 Acc	Top-4 Acc	Top-5 Acc
80.2 ± 3.0	89.3 ± 1.8	93.2 ± 0.7	95.2 ± 1.8	96.5 ± 0.3

B.4.5 Robust to Model Variations

In the main paper, all of Voyager’s experiments are conducted with gpt-4-0314. We additionally run new experiments with gpt-4-0613 and find that the performance is roughly the same (Fig. A.4). It demonstrates that Voyager is robust to model variations.

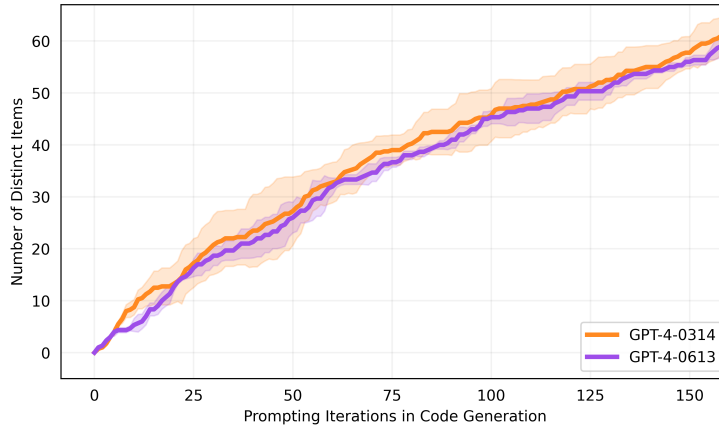


Figure A.4: VOYAGER’s performance with GPT-4-0314 and GPT-4-0613.